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Assessment - 3 CSA1709 IT

1) Intelligent Agents in a Smart Home Automation System

Introduction:

A smart home automation system uses Artificial Intelligence (AI) to control lighting, temperature, and security based on user behaviour and environmental conditions. It receives real-time information from motion sensors temperature sensors, and security devices.

The four main types of intelligent agents are:

1. Simple Reflex Agent
2. Model-Based Reflex Agent
3. Goal Based Agent
4. Utility Based Agent.

1) Simple Reflex Agent:-

Acts only on the current percept using condition-action rules. for example, it switches on a light when a motion sensors detects movement. It is fast but does not remember Past events.

2) Model-Based Reflex Agent:-

Maintains an internal state of the environment, including information that may not be directly observable. for example, it remembers that a door was locked even when the lock sensor is not being checked continuously.

3, Goal-Based Agent:-

Chooses actions that help achieve a specific goal. for example, it adjusts heating or cooling until the home reaches the user's preferred temperature of 22°c.

4, Utility-Based Agent:-

Compares possible outcomes and selects the one with the greatest overall benefit. for example, it balances comfort, electricity use, and security when deciding how to operate home devices.

Best Approach: Hybrid Intelligent Agent

A hybrid agent combining reflex model-based, goal-based and utility-based features is the most suitable choice. Reflex rules can respond immediately to danger, such as activating an alarm when a break-in is detected.

Conclusion:-

A hybrid approach is effective because it combines quick reactions with memory and careful decision-making. This makes the smart home safer, more comfortable, and more energy-efficient.

2) Unknown Maze Navigation & Search Strategies.

An autonomous agent navigating an unknown maze must discover paths, avoid obstacles, remember visited junctions and find a route to the exit without repeatedly exploring the same areas.

Uninformed Search Strategies:

- Uniform Cost Search (UCS):
Expands the node with the lowest accumulated path cost, g(n). when step costs are non-negative, it finds the least cost path. It is useful when different routes have different costs, but it may use considerable time and memory because it explores many paths.

- Iterative Deepening Search (IDS):
Runs depth-limited search repeatedly, increasing the depth-limit each time. It combines the low memory use of depth-first search with the completeness of breadth-first search. With equal steps costs, it finds a shallowest solution; however, nodes near the top may be generated repeatedly.

Agent State and Rational Decisions

A model-based agent is suitable because it maintains an internal map of the maze, recording known walls,

Open paths, and visited junctions. After each move, it updates this information using new percepts. This helps it avoids loops and the choose actions based on what it has already discovered.

Effective Combination:-

A model-based agent working with UCS is a strong choice when the aim is to find the least-cost escape route. The agent updates its map as new information becomes available, and UCS compares routes by their accumulated costs. For movement where the entire maze is not know in advance, an online search method can also be used to choose actions while exploring.

Conclusion:-

The model-based agent provides memory of the maze while the search strategy provides a systematic way to select a route. Together, they support safer exploration and a cost-effective path to the exit.